PRACTICAL GUIDELINE

Method-selection guide

Start here → Does the dataset span multiple processing routes?

IF YES — USE HIERARCHICAL BAYESIAN (ANY N)

A flat model returns one pooled slope that **misrepresents at least one subgroup**. Partial pooling recovers each route's response (here a **2.2×** FDM vs non-FDM difference) and stabilises small groups bootstrap cannot fit.

IF NO (SINGLE-ROUTE) — CHOOSE BY RISK POSTURE

Bootstrap

COST-SENSITIVE · SHARPER INTERVALS

- \+ Tightest widths (25.9 MPa) and ~350× cheaper per fit
- \+ Equivalent score to Bayesian above $n \approx 20$
- – Overconfident: under-covers (PICP 0.846) on hard held-out points

Bayesian NUTS

SAFETY-CRITICAL · COVERAGE GUARANTEE

- \+ Prior-stabilised coverage holds at the smallest n
- \+ Conservative intervals (PICP 1.000) for qualification margins
- – Wider (28.9 MPa) and ~350× the compute

Default: above $n \approx 20$ the flat methods converge in coverage and width — prefer bootstrap on cost unless a conservative coverage guarantee is required. The choice is a **risk-posture** decision, not a performance ranking.